

	1	2	3	4	5	6	7	8
A	OPENDUAL READER Dual-frequency mullion reader / engineering prototype A0 Nominal 12 V / ESP32-WROOM-32E / HID Prox + EM4100 LF / ISO14443A HF							
B	01 / POWER AND PROTECTION File: power.kicad_sch				02 / ESP32 AND SERVICE File: mcu.kicad_sch			
C	03 / LF AND HF MODULES File: rf.kicad_sch				04 / OSDP AND WIEGAND File: panel.kicad_sch			
D								
E	05 / STATUS, TAMPER, EXPANSION File: ui.kicad_sch							
F	REVIEW HOLD: RF coexistence, sample dimensions, credentials and panel interoperability untested. See docs/open-issues.md. This project is not released for fabrication.						OpenDual / open hardware Sheet: / File: OpenDual-Reader.kicad_sch Title: OpenDual Reader / independent prototype Size: A3Date:KiCad E.D.A. 10.0.6 Rev: A0Id: 1/6	
	1	2	3	4	5	6	7	8

01 / POWER AND PROTECTION

A

B

C

D

E

F

A

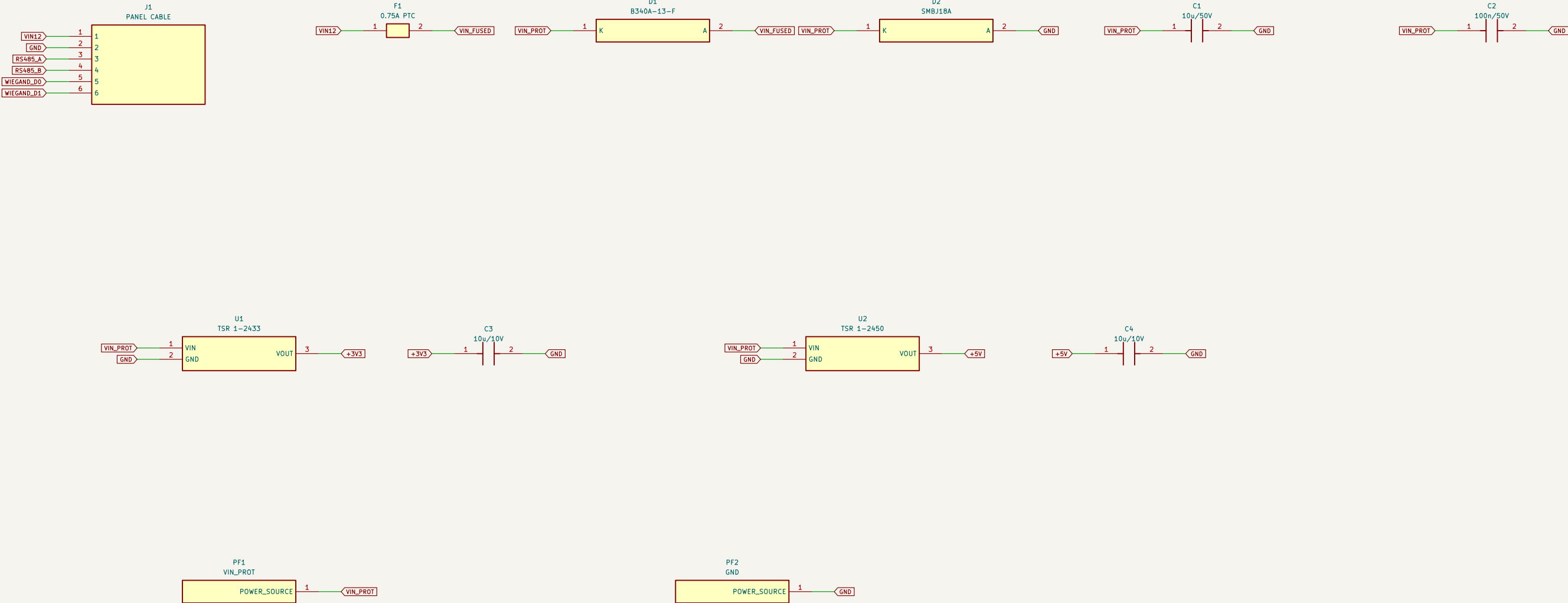
B

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D

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F



INPUT: 9..16 VDC, nominal12V. Bench target0..50C pending sealed–enclosure thermal testing.

B340A blocks reversed supply. Fuse before TVS: transient coordination requires surge testing.

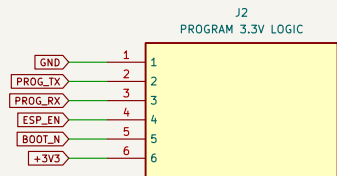
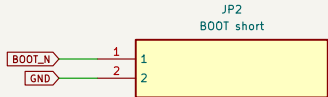
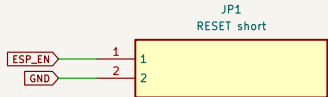
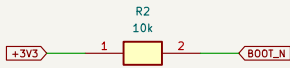
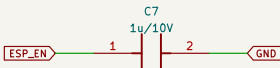
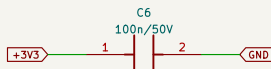
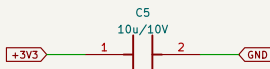
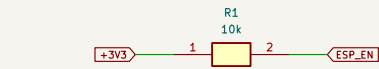
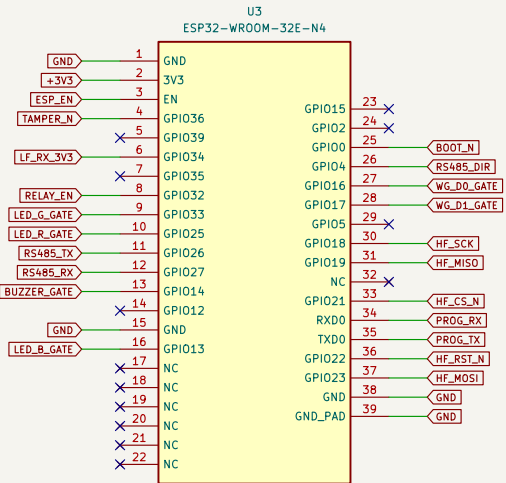
Separate 1A rails: 3.43W load allocation; approximately 4.6W at input. Thermal testing required.

No USB port. Programming connector3V3 is a measurement point only. No backfeed path permitted.

Sheet: /01 / POWER AND PROTECTION/
File: power.kicad_sch

Title: 01 / POWER AND PROTECTION

Size: A3	Date:	Rev: A0
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GPIO0 boot: short JP2 then pulse JP1. Use 3,3V UART adapter and common ground.

Strapping GPIO2/5/12/15 unused; flash/NC pads not used. GPIO34/36 have external functions only.

EN 10k + 1u startup delay. Actuator pull-downs: 100k except GPIO14 buzzer base uses 2.2k.

U3 module pad39 GND must be soldered. RF/copper clearance follows board rule areas.

Sheet: /02 / ESP32 AND SERVICE/
File: mcu.kicad_sch

Title: 02 / ESP32 AND SERVICE

Size: A3

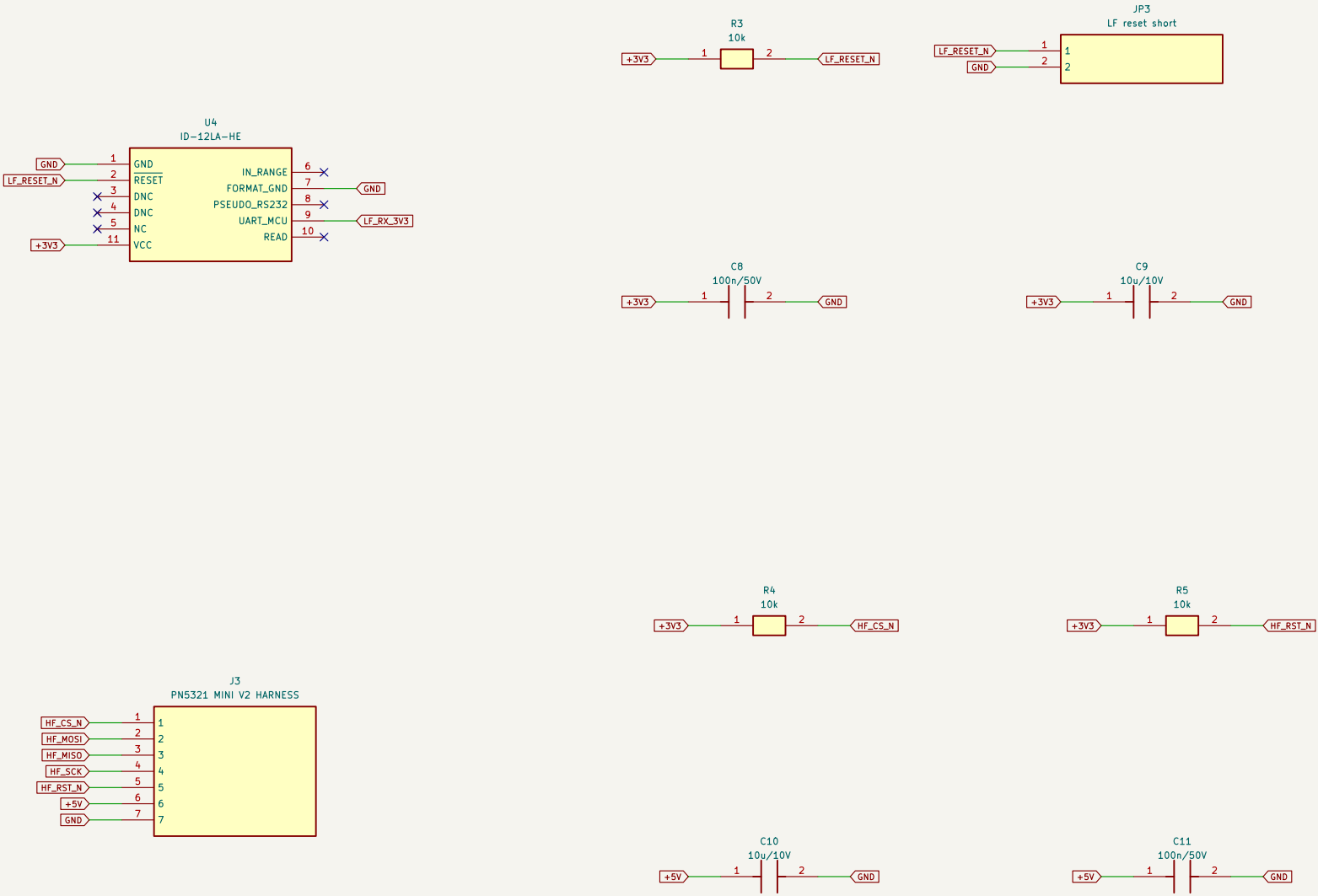
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Rev: A0

KiCad E.D.A. 10.0.6

Id: 3/6

03 / LF AND HF MODULES



LF: 125kHz EM4100-compatible + supported HID Prox subset; closed ID Innovations decoder.

LF raw payload kept intact; sample evidence required for bit ordering and wristband format.

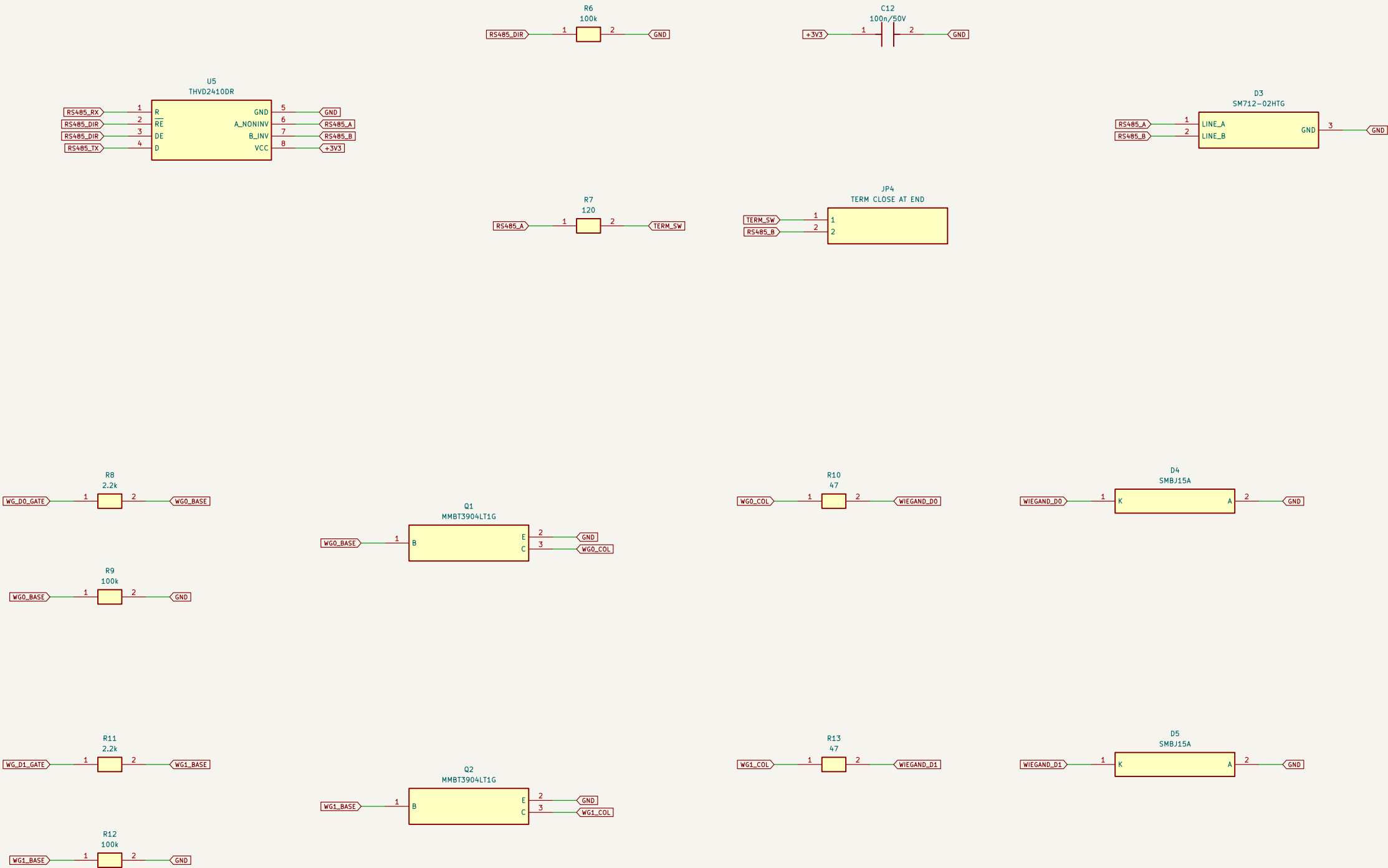
HF: ELECHOUSE PN5321 MINI V2 SKU NFC_EX_MINI; +5V supply, 3.3V SPI; exact supplied harness.

SPI mode0 LSB-first. IRQ not exposed; status polling. Reset open-drain. Bridge SPI selector only.

RF modules include matching. No invented antenna L/C values. Neither antenna is stacked.

No auto LF field control in A0. If simultaneous operation fails, GPIO/interleaving ECO is required.

04 / OSDP AND WIEGAND



OSDP Peripheral Device only. A pin6 is NONINVERTING; B pin7 is INVERTING. Share GND.

JP4 default open; close 120ohm termination only at cable end. Bias supplied once at control panel.

THVD2410 has failsafe reception and +/-70V bus fault protection. SM712 needs short ground return.

Wiegand: external pull-up 0..12V nominal (15V max), sink 5mA max; low target <=0.5V.

GPIO HIGH sinks line. All drivers off at reset; panel voltage never connects to MCU pins.

Connector-level IEC surge/ESD performance is untested. This is a nonisolated interface.

05 / STATUS, TAMPER, EXPANSION

Active-high drivers; remote tamper; logic-only expansion. See docs/architecture.md.

Remote tamper: J4.1 -> RT1 1k -> SW1 KSC223G LFG (NO) -> J4.2 GND. Pressed = closed; released = tamper.

